

TASK BASED ON MOVIE SUCCESS ANALYSIS

Program:

```
# TASKS (Using Movie Success Analysis)

import pandas as pd

df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip") # Change filename if
needed

print(df.shape)

print(df.columns.tolist())
```

Output:

```
(7668, 15)

['name', 'rating', 'genre', 'year', 'released', 'score', 'votes', 'director', 'writer', 'star', 'country',
'budget', 'gross', 'company', 'runtime']
```

Program:

```
# TASKS (Using Movie Success Analysis(Task 1: Budget vs Rating))

# Import Libraries

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Load Dataset

df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip")

# Display First 5 Records

print(df.head())
```

```

# Check Missing Values
print(df[['budget','score']].isnull().sum())

# Remove Missing Values
df = df.dropna(subset=['budget','score'])

# Scatter Plot
plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x='budget', y='score')

plt.title("Budget vs Movie Rating")
plt.xlabel("Budget")
plt.ylabel("IMDb Score")
plt.show()

```

Output:

	name	rating	genre	year \
0	The Shining	R	Drama	1980
1	The Blue Lagoon	R	Adventure	1980
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980
3	Airplane!	PG	Comedy	1980
4	Caddyshack	R	Comedy	1980

	released	score	votes	director \
0	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick
1	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser
2	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner
3	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams
4	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis

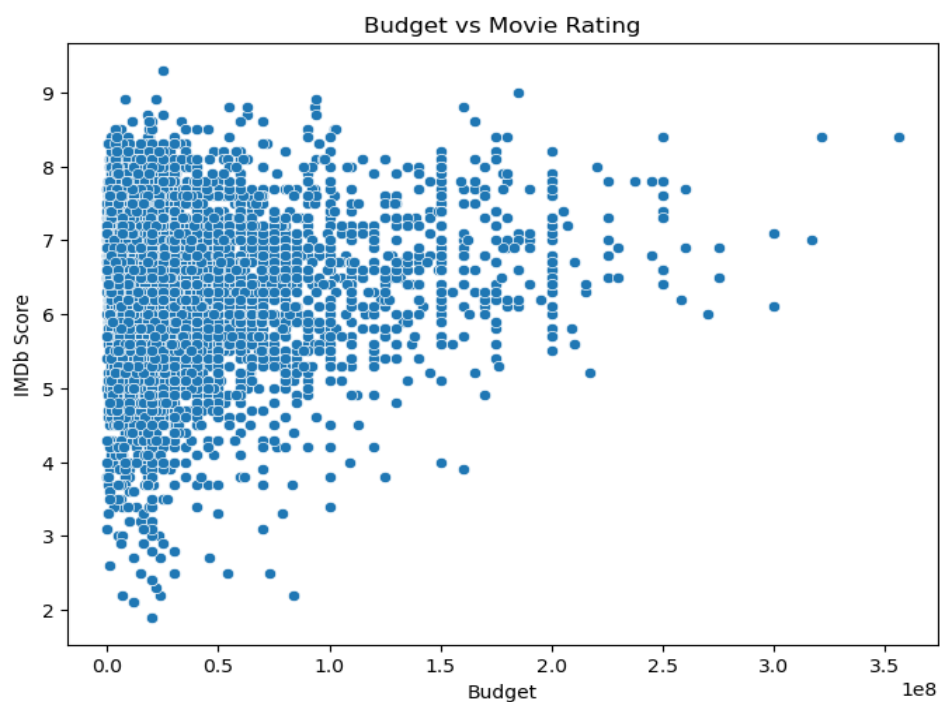
	writer	star	country	budget \
0	Stephen King	Jack Nicholson	United Kingdom	19000000.0
1	Henry De Vere Stacpoole	Brooke Shields	United States	4500000.0
2	Leigh Brackett	Mark Hamill	United States	18000000.0
3	Jim Abrahams	Robert Hays	United States	3500000.0
4	Brian Doyle-Murray	Chevy Chase	United States	6000000.0

	gross	company	runtime
0	46998772.0	Warner Bros.	146.0
1	58853106.0	Columbia Pictures	104.0
2	538375067.0	Lucasfilm	124.0
3	83453539.0	Paramount Pictures	88.0
4	39846344.0	Orion Pictures	98.0

budget 2171

score 3

dtype: int64



Program:

```
# TASKS (Using Movie Success Analysis (Task 2: Genre Popularity))
```

```
# Import Libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load Dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip")
```

```
# Count Genres
```

```
genre_count = df['genre'].value_counts()
```

```
print(genre_count)
```

```
# Plot
```

```
plt.figure(figsize=(10,6))
```

```
genre_count.plot(kind='bar')
```

```
plt.title("Genre Popularity")
```

```
plt.xlabel("Genre")
```

```
plt.ylabel("Number of Movies")
```

```
plt.show()
```

Output:

```
genre
```

```
Comedy    2245
```

```
Action    1705
```

```
Drama     1518
```

```
Crime      551
```

```
Biography  443
```

```
Adventure  427
```

Animation 338

Horror 322

Fantasy 44

Mystery 20

Thriller 16

Family 11

Sci-Fi 10

Romance 10

Western 3

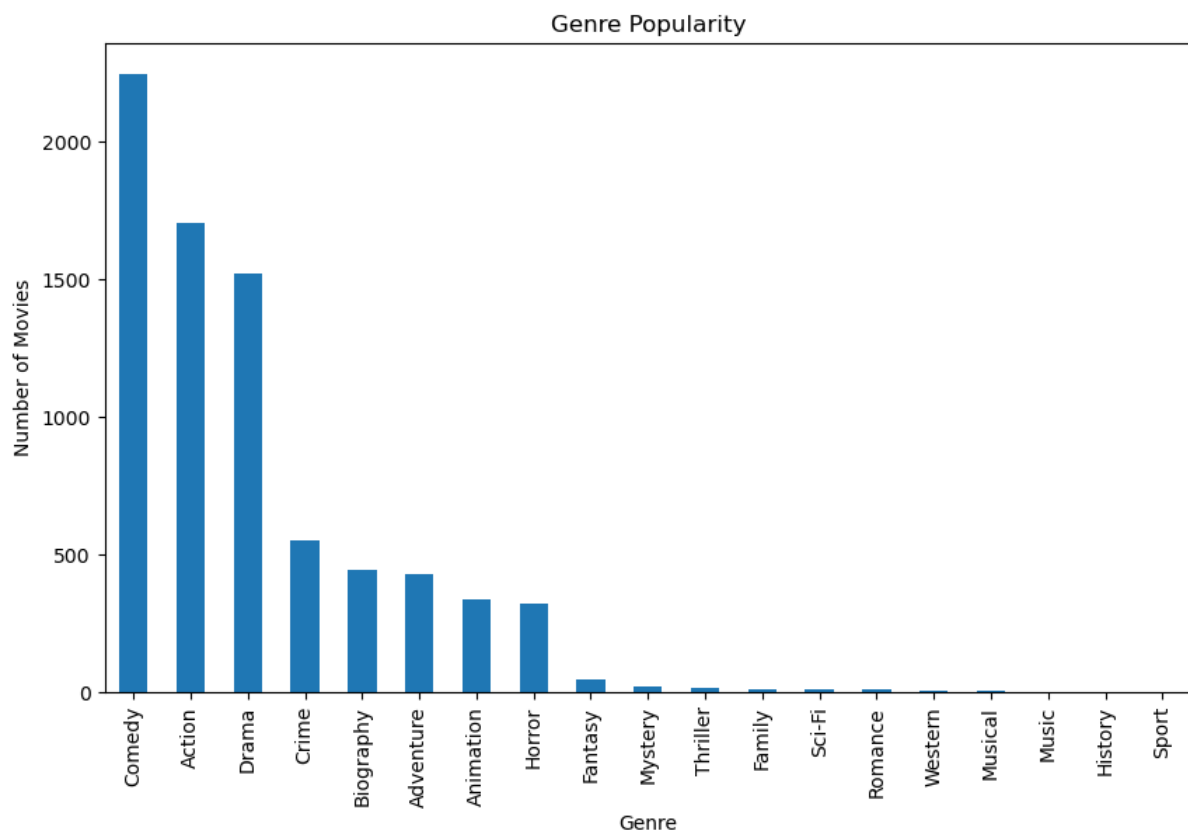
Musical 2

Music 1

History 1

Sport 1

Name: count, dtype: int64



Program:

```
# TASKS (Using Movie Success Analysis (Task 3: Director Performance))
```

```
# Import Libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load Dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip")
```

```
# Average Rating of Directors
```

```
director = df.groupby('director')['score'].mean()
```

```
# Top 10 Directors
```

```
top_director = director.sort_values(ascending=False).head(10)
```

```
print(top_director)
```

```
# Plot
```

```
plt.figure(figsize=(10,6))
```

```
top_director.plot(kind='bar')
```

```
plt.title("Top 10 Directors by Average Rating")
```

```
plt.xlabel("Director")
```

```
plt.ylabel("Average IMDb Score")
```

```
plt.show()
```

Output:

```
director
```

```
Olivier Nakache      8.50
```

```
Anurag Kashyap      8.50
```

Marco Tullio Giordana 8.50

Aamir Khan 8.40

Sergio Leone 8.40

Bob Persichetti 8.40

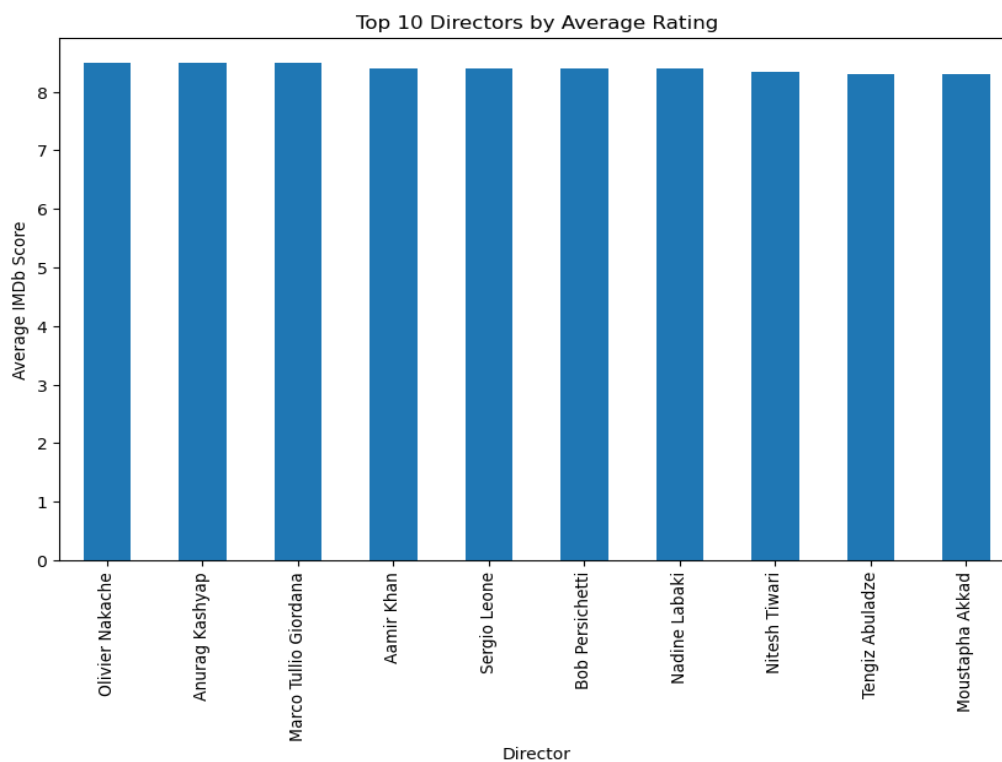
Nadine Labaki 8.40

Nitesh Tiwari 8.35

Tengiz Abuladze 8.30

Moustapha Akkad 8.30

Name: score, dtype: float64



Program:

TASKS (Using Movie Success Analysis(Task 4: Revenue Trends))

Import Libraries

import pandas as pd

import matplotlib.pyplot as plt

```
# Load Dataset

df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip")


# Group Revenue by Year

revenue = df.groupby('year')['gross'].sum()

print(revenue)


# Plot

plt.figure(figsize=(10,6))

revenue.plot()

plt.title("Revenue Trend Over Years")

plt.xlabel("Year")

plt.ylabel("Total Revenue")

plt.grid(True)

plt.show()
```

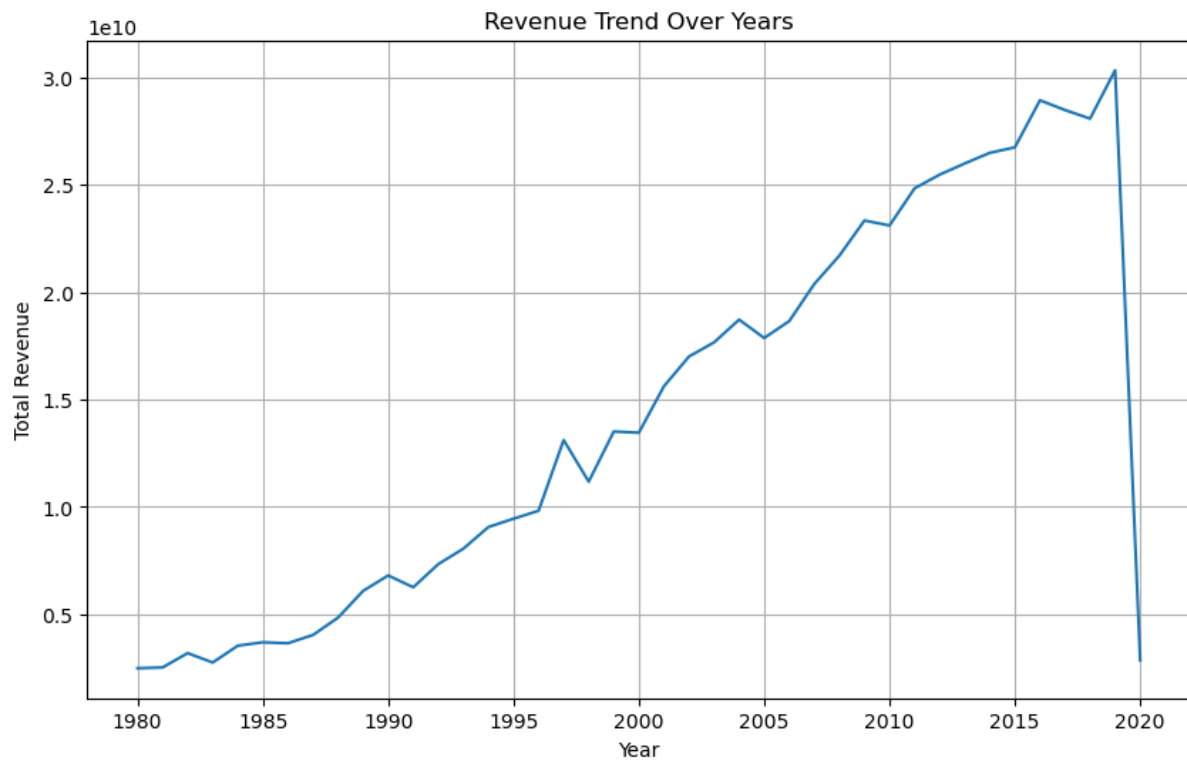
Output:

```
year
1980  2.483667e+09
1981  2.520065e+09
1982  3.187581e+09
1983  2.750005e+09
1984  3.533092e+09
1985  3.691509e+09
1986  3.647512e+09
1987  4.033434e+09
1988  4.840386e+09
1989  6.087251e+09
1990  6.804236e+09
```


1991	6.254771e+09
1992	7.334168e+09
1993	8.053959e+09
1994	9.062926e+09
1995	9.448305e+09
1996	9.819528e+09
1997	1.311228e+10
1998	1.117602e+10
1999	1.350964e+10
2000	1.345413e+10
2001	1.561763e+10
2002	1.700344e+10
2003	1.766193e+10
2004	1.872175e+10
2005	1.786661e+10
2006	1.865407e+10
2007	2.038367e+10
2008	2.170216e+10
2009	2.333868e+10
2010	2.310660e+10
2011	2.483769e+10
2012	2.547619e+10
2013	2.599585e+10
2014	2.649148e+10
2015	2.674601e+10
2016	2.893884e+10
2017	2.848525e+10
2018	2.808529e+10
2019	3.033965e+10

2020 2.848299e+09

Name: gross, dtype: float64



Program:

```
# TASKS (Using Movie Success Analysis(Task 5: Movie Recommendation Insights))
```

```
# Import Libraries
```

```
import pandas as pd
```

```
# Load Dataset
```

```
df = pd.read_csv(r"C:\Users\Alana\Downloads\movies.csv.zip")
```

```
# Remove Missing Values
```

```
df = df.dropna(subset=['score','gross','budget'])
```

```
# Recommend Movies
```

```
recommend = df[
```

```

(df['score'] >= 8.0) &
(df['gross'] >= 100000000) &
(df['budget'] >= 50000000)
]
print("Recommended Movies")
print(recommend[['name',
                  'genre',
                  'director',
                  'score',
                  'budget',
                  'gross']])

```

Output:

Recommended Movies

	name	genre	director	score \
1843	Terminator 2: Judgment Day	Action	James Cameron	8.5
2245	Jurassic Park	Action	Steven Spielberg	8.1
2445	Forrest Gump	Drama	Robert Zemeckis	8.8
2643	Heat	Crime	Michael Mann	8.2
2646	Braveheart	Biography	Mel Gibson	8.3
...	...	...	...	...
7252	Spider-Man: Into the Spider-Verse	Animation	Bob Persichetti	8.4
7445	Avengers: Endgame	Action	Anthony Russo	8.4
7450	Joker	Crime	Todd Phillips	8.4
7462	1917	Drama	Sam Mendes	8.3
7464	Ford v Ferrari	Action	James Mangold	8.1

	budget	gross
1843	102000000.0	5.208812e+08
2245	63000000.0	1.033928e+09
2445	55000000.0	6.782261e+08
2643	60000000.0	1.874368e+08
2646	72000000.0	2.132162e+08
...	...	...
7252	90000000.0	3.755408e+08
7445	356000000.0	2.797501e+09
7450	55000000.0	1.074427e+09
7462	95000000.0	3.846236e+08
7464	97600000.0	2.255082e+08

[62 rows x 6 columns]